
Stakeholder Engagement Plan

Zimbabwe Agricultural Extension RAG System

1. Stakeholder Identification

Stakeholder Group 1: Women Smallholder Farmers & Women's Farming Cooperatives

Who they are: Members of women-led farming cooperatives, savings groups, and informal farmer associations across Zimbabwe's communal and resettlement farming areas. Women carry out a large share of day-to-day crop production and household food provisioning, though they frequently have less formal land ownership and less direct access to extension services than men.

Why they are important to the project: Women farmers are among the primary intended end users of the system, and are also one of the groups most likely to be underrepresented in the existing extension literature the dataset is built from. Their engagement is essential to the project's stated value of inclusivity, and to ensuring the tool doesn't simply replicate existing gaps in who receives agricultural support.

What matters most to them: Practical, trustworthy advice that fits their real day-to-day farming responsibilities; safe access to information without needing to navigate systems designed around male heads of household; tools that don't assume high literacy, English fluency, or unrestricted smartphone access; and recognition of their existing knowledge and labour rather than being treated as passive recipients of expert advice.

Stakeholder Group 2: AGRITEX Extension Officers

Who they are: Government field officers employed under Zimbabwe's Department of Agricultural, Technical and Extension Services, responsible for delivering in-person agronomic advice, running demonstrations, and acting as the trusted local link between the Ministry of Agriculture and farming communities.

Why they are important to the project: AGRITEX officers are both a primary content source for the dataset and the professional group whose day-to-day role is most directly affected by the system. Their buy-in determines whether the tool is trusted and adopted by farmers, and their domain expertise is needed to validate retrieval quality on an ongoing basis.

What matters most to them: That the system supports rather than undermines their professional role and credibility; that content attributed to AGRITEX is accurate, since errors could damage institutional trust; and that they are consulted as content authorities rather than bypassed by a tool built without their input.

2. Potential Benefits and Risks

Women Smallholder Farmers & Women's Farming Cooperatives

Potential benefits:

- Faster access to reliable agricultural advice without needing to wait for or travel to a male-oriented extension visit.
- Greater agricultural autonomy and confidence in decision-making around crops, pests, and soil management.
- Improved household food security and income stability from better-informed farming decisions.
- A channel through which women's farming priorities and questions can be reflected in the knowledge base over time.

Potential harms or concerns:

- Exclusion if the system assumes smartphone ownership, data access, English literacy, or independent decision-making that some women may not have.
- Reinforcement of gender stereotypes if source content implicitly frames the 'farmer' as male or overlooks tasks predominantly carried out by women.
- Underrepresentation in the underlying dataset, since much formal extension literature has historically centred male-headed households.

Ethical, social, cultural, and economic considerations:

In some communities, cultural norms around household decision-making or device access may limit how freely women can use a digital tool independently, so engagement needs to be sensitive to existing social structures rather than assuming equal access. Economically, data and device costs can be a real barrier. There is also an ethical responsibility to avoid extracting women's input to legitimise the project without genuinely acting on it.

AGRITEX Extension Officers

Potential benefits:

- Reduced burden of repetitive, routine queries, freeing time for complex cases and field visits.
- A tool that extends the reach of AGRITEX-approved advice into areas officers cannot regularly visit.
- A structured channel to contribute and continually refine the content farmers rely on.

Potential harms or concerns:

- Concern about role displacement or de-skilling if the tool is perceived as a replacement rather than a complement to their work.
- Reputational risk if inaccurate or outdated advice is retrieved under the AGRITEX name.
- Disagreement over who has authority to approve or update content in the knowledge base.

Ethical, social, cultural, and economic considerations:

There are direct livelihood and professional-identity implications if officers feel the tool competes with rather than supports their role. Institutionally, AGRITEX's accountability for the accuracy of publicly attributed advice raises questions of liability and oversight that need to be resolved collaboratively rather than assumed by the project team.

3. Engagement Method

Women Smallholder Farmers & Women's Farming Cooperatives — Distributed Dialogue

Distributed Dialogue, as catalogued in the Action Catalogue, is a decentralised engagement method that gathers input from many geographically dispersed participants without requiring them to travel to a single central event. This method is appropriate for women's farming cooperatives because it works with existing social infrastructure rather than asking women to take on additional travel, time, or cost burdens to participate — barriers that disproportionately affect women balancing farming, household, and caregiving responsibilities. Rather than requiring a formal, centralised workshop that may be difficult for rural participants to attend, distributed dialogue allows engagement to happen where women already gather, respecting their time and existing community structures.

In practice, the engagement would be conducted by working with local partner organisations and AGRITEX's community networks to identify existing women's cooperative and savings-group meetings across a sample of Zimbabwe's Natural Regions. Trained local facilitators — ideally women from or known to the community, to encourage open participation — would join these regular meetings with a short, structured discussion guide covering: what agricultural questions come up most often, what has made past advice or tools hard to use, and what would make a new digital tool trustworthy and genuinely useful. Sessions would be conducted in Shona or Ndebele as appropriate, with facilitator notes (and audio recordings where consented to) collected and synthesised centrally by the project team. Running this across multiple regions and multiple rounds — an initial round to inform design, and a follow-up round once a prototype exists — allows the method to capture a genuinely dispersed and locally grounded picture of priorities, rather than the views of a single, more accessible group standing in for the whole community.

AGRITEX Extension Officers — User Committee

A User Committee is appropriate for AGRITEX officers because their relationship to the project is ongoing and professional rather than a one-off consultation: they are both a content source and a continuing quality-assurance authority for the system. Unlike a single workshop, a standing committee gives officers a structured, recurring role with real influence over how the system evolves, which directly addresses their concern about being consulted as authorities rather than bypassed.

This method is suitable for this group specifically because officers already operate within an institutional hierarchy and professional review culture — a formal committee structure fits naturally alongside their existing AGRITEX reporting and quality-assurance practices, more so than an informal or one-off dialogue format would. In practice, the engagement would involve forming a small standing committee of AGRITEX officers, ideally representing different provinces and Natural Regions, that meets quarterly. At each session, the committee would review a sample of real farmer queries and the documents the system retrieved in response, flag any inaccurate, outdated, or missing content, and discuss emerging issues such as new pest outbreaks or seasonal advisories that should be added to the knowledge base. The committee would also hold a formal sign-off role before major content or ranking changes are deployed, giving officers genuine decision-making authority rather than a purely advisory one. Over time, this structure builds a continuous feedback loop between the people using AGRITEX's institutional knowledge day-to-day and the system meant to extend that knowledge digitally.

4. Influence on Project Design

Feedback from both stakeholder groups is intended to shape the project across several concrete dimensions:

Dataset:

Distributed dialogue input from women's cooperatives will help identify topics and crops that are under-covered in current formal literature but matter most to women farmers, guiding what content is prioritised for collection or translation. AGRITEX committee review will directly flag gaps, outdated material, or inaccuracies to correct in the knowledge base.

Model design:

Retrieval ranking can be adjusted to weight officer-validated, high-confidence sources more heavily, and low-confidence or low-agreement retrievals can be routed for committee review before being trusted in production.

Outputs:

Based on dialogue feedback, outputs may need simplified language options, local-language responses, or audio-based delivery to reduce literacy and language barriers for women farmers with limited access to formal English-language extension materials.

Evaluation:

Relevance labelling for nDCG@5 evaluation can be expanded beyond central domain experts to include periodic input from the AGRITEX user committee and structured feedback from women's cooperatives, so evaluation reflects real user needs and not only expert judgement.

Deployment decisions:

Findings may support a phased rollout that pairs digital access with AGRITEX officers as intermediaries in areas with lower connectivity or digital literacy, and prioritise offline-capable or audio-first delivery channels identified as important during engagement with women's groups.

References

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